Brain Tumor Segmentation using multi scale attention U-Net with EfficientNetB4 encoder for enhanced MRI analysis

A report submitted in partial fulfillment of the requirements
for the award of the Degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE ENGINEERING

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CERTIFICATE OF COMPLETION

This is to certify that the work entitled **"Brain Tumor Segmentation using multi scale attention U-Net with EfficientNetB4 encoder for enhanced MRI analysis"** is the bonafide work of **P Chandu, M Suryam, G Navaneeth, S Manoj** carried out under my guidance and supervision for the Minor Project of the Bachelor of Technology in the Department of Computer Science and Engineering under RGUKT Nuzvid. This work was done during the academic session December 2025 – April 2026 under our guidance.

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CERTIFICATE OF EXAMINATION

This is to certify that the work entitled **"Brain Tumor Segmentation using multi scale attention U-Net with EfficientNetB4 encoder for enhanced MRI analysis"** is the bonafide work of **P Chandu, G Navaneeth, M Suryam,S Manoj**, and hereby accord our approval of it as a study carried out and presented in a manner required for its acceptance in the Minor Project of Bachelor of Technology for which it has been submitted. This approval does not necessarily endorse or accept every statement made, opinion expressed or conclusion drawn, as recorded in this report. It only signifies the acceptance of this report for the purpose for which it has been submitted.

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DECLARATION

We **P Chandu, G Navaneeth, M Suryam,S Manoj**, hereby declare that the project report submitted for partial fulfillment of the requirements for the award of the Bachelor of Technology degree in Computer Science Engineering of the Rajiv Gandhi University of Knowledge Technologies, Nuzvid, is a bonafide work done by us under the guidance of **Mr. Jaya Krishna**. This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the university and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously been formed as the basis for the award of any degree, diploma, or similar title from any other university.

Date: _____

Place: _____

P Chandu G Navaneeth

M Suryam S Manoj

Acknowledgments

I express sincere gratitude to my guide, faculty, and peers for their support during this work. I also acknowledge the open-source ecosystem, especially PyTorch, Albumentations, and pretrained EfficientNet resources used in this project.

Abstract

Accurate brain tumor segmentation from MRI scans is a critical step in computer-aided diagnosis. This report presents a hybrid deep learning pipeline that combines preprocessing, EfficientNet-based encoding, custom Inception-Dense and depthwise-separable modules, attention mechanisms, and a decoder for pixel-level mask prediction. The implementation was trained in multiple phases, including experiments with and without ViT-assisted components, and optimized using Dice/Tversky-oriented objectives. The best run in the notebook reaches a final IoU near 0.847 and precision/recall above 0.91 in one training regime, while a stabilized validation-centric regime reaches peak validation Mean IoU around 0.726 with balanced precision, recall, and specificity. These outcomes show strong segmentation quality and practical robustness for medical image workflows.

Keywords Used: Brain Tumor Segmentation, Medical Imaging, Deep Learning, EfficientNet, Hybrid Encoder-Decoder, IoU, MRI.

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Chapter 1

Introduction

Brain tumors can be life-threatening when not detected and localized at an early stage. Manual segmentation of tumor regions in MRI scans is time-consuming and depends heavily on clinical expertise. Automated segmentation using deep neural models reduces variability and supports faster medical analysis.

This project focuses on building a segmentation model that learns robust spatial features from MRI data by combining:

- image enhancement and mask-aligned preprocessing,
- pretrained EfficientNet feature extraction,
- custom encoder/decoder design for dense prediction,
- attention and hybrid feature fusion for better localization.

Chapter 2

Objectives

- Develop an end-to-end tumor segmentation pipeline from MRI image to binary/region mask output.
- Improve learning stability through preprocessing, augmentation, and staged training.
- Evaluate segmentation quality using IoU, precision, recall, and specificity.
- Compare training behavior across multiple model configurations.

Chapter 3

Related Work

Classical segmentation methods use thresholding, region growing, and edge cues, but these methods struggle with heterogeneous tumor shapes and intensities. Modern CNN-based approaches such as U-Net families improved medical segmentation significantly. More recent work combines pretrained backbones (EfficientNet, ResNet variants), attention modules, and transformer-like global context modeling.

In this project, the implemented design follows this trend by combining a pretrained convolutional backbone with custom multi-scale modules and attention-enhanced decoding.

Chapter 4

Methodology

4.1 Notebook Pipeline Overview

The source implementation in `Final_Tumor_code.ipynb` includes:

- data loading and visualization,
- preprocessing (sharpening, edge enhancement, CLAHE-like contrast operations),
- transform pipelines with resizing and normalization,
- dataset and dataloader construction,
- staged model training and metric tracking.

4.2 Data and Augmentation

The code uses custom dataset handling with Albumentations-based transforms. Common operations include resizing to fixed resolution (e.g., 224 or 256), normalization using ImageNet-style means/stds, and paired image-mask augmentation.

4.3 Model Architecture

From notebook classes and blocks, the implemented architecture contains:

- **Backbone:** EfficientNet-B4 feature extractor (loaded/fine-tuned in stages).
- **Encoder:** custom block combining extracted hierarchical features.
- **Feature modules:** InceptionDense and depthwise-separable components for efficient multi-scale representation.
- **Attention:** channel/spatial weighting to improve tumor region focus.
- **Decoder:** segmentation head that progressively reconstructs mask predictions.

4.4 Losses and Optimization

The notebook defines Dice/Tversky-inspired losses and combined objectives. Optimizers include Adam and AdamW with learning-rate scheduling behavior in later training variants.

4.5 Detailed Project Flow

To provide reproducibility and publication-ready documentation, two flowcharts were created in Draw.io and stored in:

- `report_assets/diagrams/brain_tumor_flowcharts.drawio`

The diagrams contain:

- end-to-end processing pipeline (from MRI input to evaluation),
- architecture-level view (EfficientNet encoder, hybrid blocks, decoder, segmentation head).

Chapter 5

Experimental Setup

5.1 Software Environment

- Python with PyTorch and Torchvision
- Albumentations for augmentation
- Matplotlib and NumPy for analysis and visualization

5.2 Training Configuration (From Notebook Logs)

- Batch size observed in dataloaders: 8
- Epochs in major runs: 100
- Optimizers: Adam / AdamW
- Metrics: IoU (or Mean IoU), Precision, Recall, Specificity, Loss

Chapter 6

Results and Analysis

6.1 Quantitative Results From Saved Outputs

Two key training regimes visible in notebook outputs are summarized in Table 6.1.

Table 6.1: Summary of key training runs captured in notebook outputs

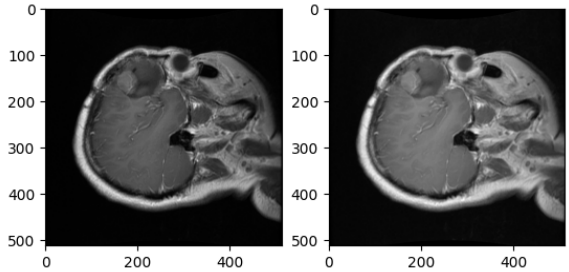
Run	Best/Final Snapshot	Notes
Run-A (Cell 110 logs)	Best IoU $\approx$ 0.8475 at Epoch 99; Final IoU $\approx$ 0.8467 at Epoch 100; Final Precision/Recall $\approx$ 0.9139/0.9137	Strong convergence trend from very low initial IoU (0.0734) to high-quality segmentation.
Run-B (Cell 116 logs)	Best Validation Mean IoU $\approx$ 0.7263 at Epoch 45; Final Validation Mean IoU $\approx$ 0.7236 at Epoch 100; Final Val Precision/Recall/Specificity $\approx$ 0.8412/0.8921/0.9338	More stability-focused setup with explicit validation tracking and balanced metrics.

6.2 Training Behavior

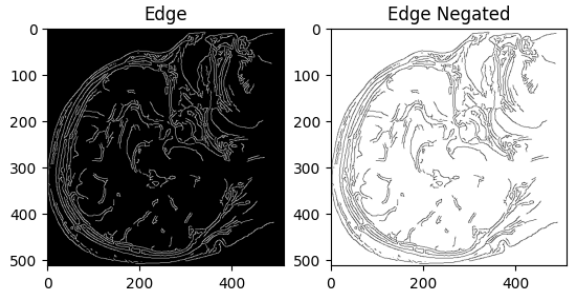
The first regime demonstrates strong end-stage overlap and class discrimination. The second regime provides smoother validation monitoring and better generalization diagnostics. Together, they indicate that the architecture can be tuned either for aggressive overlap optimization or validation-balanced performance.

6.3 Qualitative Output Samples

Representative image outputs extracted from the notebook are shown below.

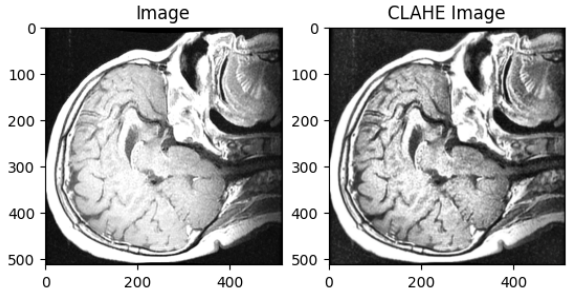


(a) Input and sharpening comparison

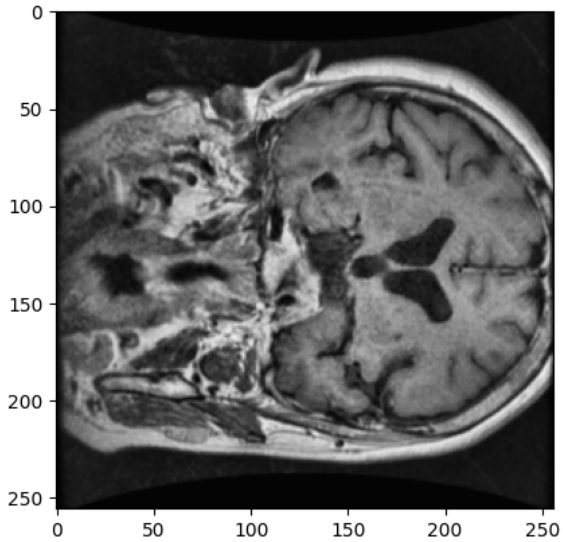


(b) Edge extraction

Figure 6.1: Early preprocessing stages

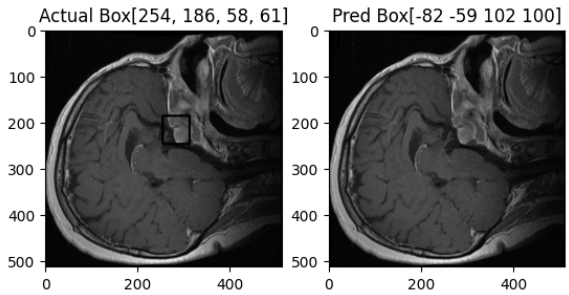


(a) Contrast-enhanced image

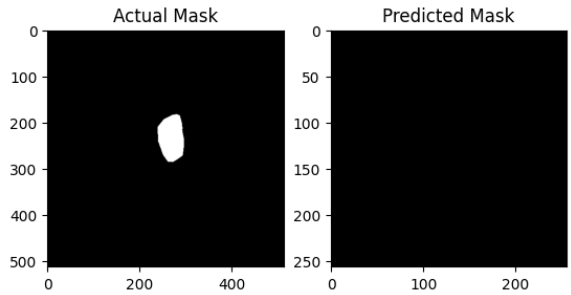


(b) Augmented sample

Figure 6.2: Preprocessed and augmented training inputs

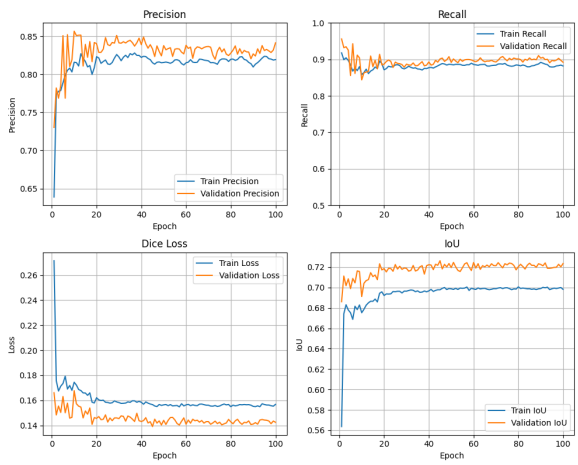


(a) Bounding/localization style output

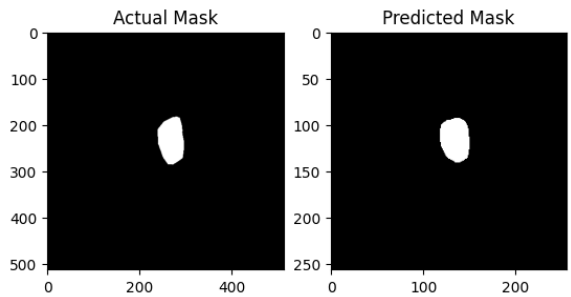


(b) Actual mask visualization

Figure 6.3: Mask-oriented outputs



(a) Metric curves

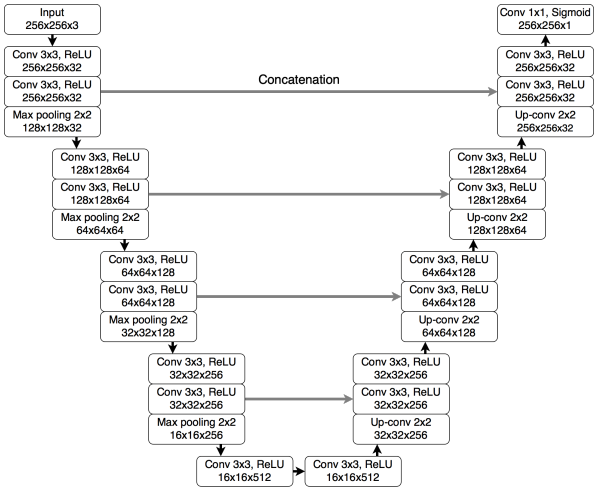


(b) Predicted vs mask sample

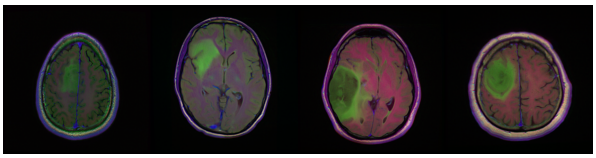
Figure 6.4: Training dynamics and segmentation output

6.4 Architecture and External Reference Visuals

This section adds architecture-related visuals and external reference figures relevant to MRI tumor segmentation.



(a) Reference U-Net style architecture



(b) Sample MRI + mask visualization

Figure 6.5: Public online images related to brain MRI segmentation

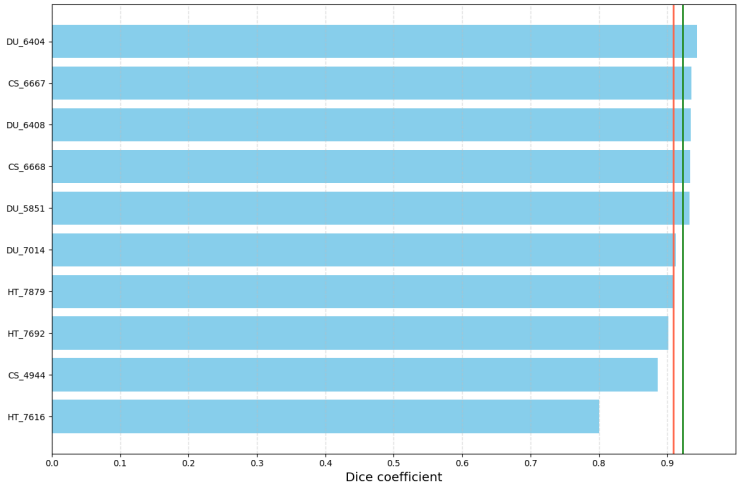
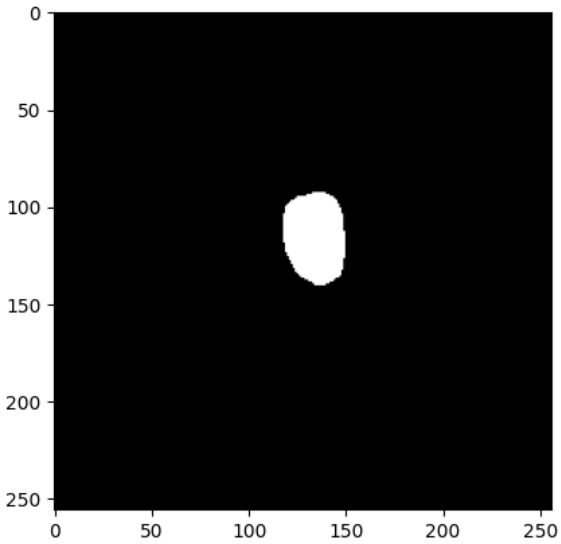
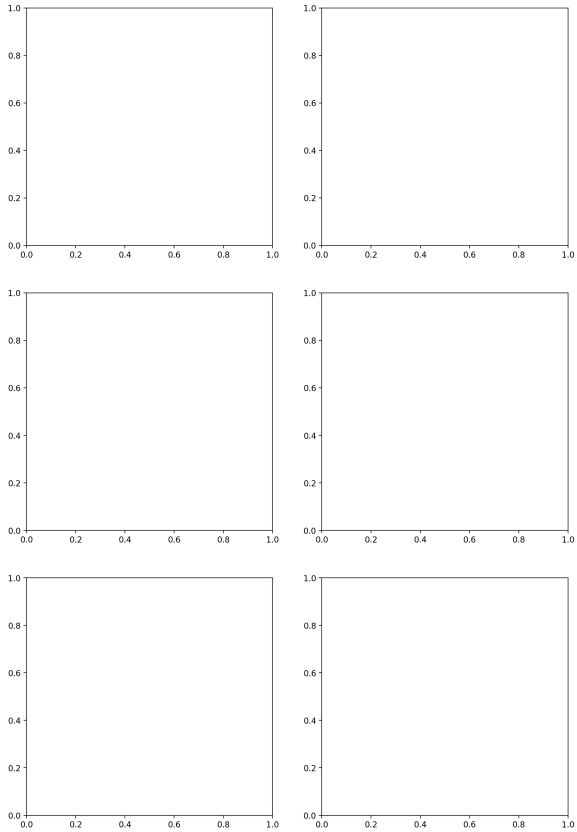


Figure 6.6: Reference segmentation performance distribution (DSC)



(a) Model-predicted mask output



(b) Multi-sample output grid

Figure 6.7: Architecture outcome visuals from your notebook outputs

Chapter 7

Conclusion

This work demonstrates a practical MRI brain tumor segmentation pipeline built from a hybrid deep architecture around EfficientNet features and custom decoder modules. The recorded notebook outputs show substantial performance gains across training epochs, reaching high IoU in one run and consistent validation behavior in another.

The resulting workflow is suitable as a baseline for advanced clinical-grade segmentation studies.

Chapter 8

Future Scope

- Integrate k-fold cross-validation for stronger reliability estimates.
- Add post-processing (CRF/morphology) for boundary refinement.
- Evaluate on multi-institution MRI datasets for domain shift analysis.
- Extend to multi-class tumor subregion segmentation.

Chapter 9

References

1. M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," *ICML*, 2019.

2. O. Ronneberger *et al.*, "U-Net: Convolutional Networks for Biomedical Image Segmentation," *MICCAI*, 2015.

3. A. Paszke *et al.*, "PyTorch: An Imperative Style, High-Performance Deep Learning Library," *NeurIPS*, 2019.

4. A. Buslaev *et al.*, "Albumentations: Fast and Flexible Image Augmentations," *Information*, 2020.

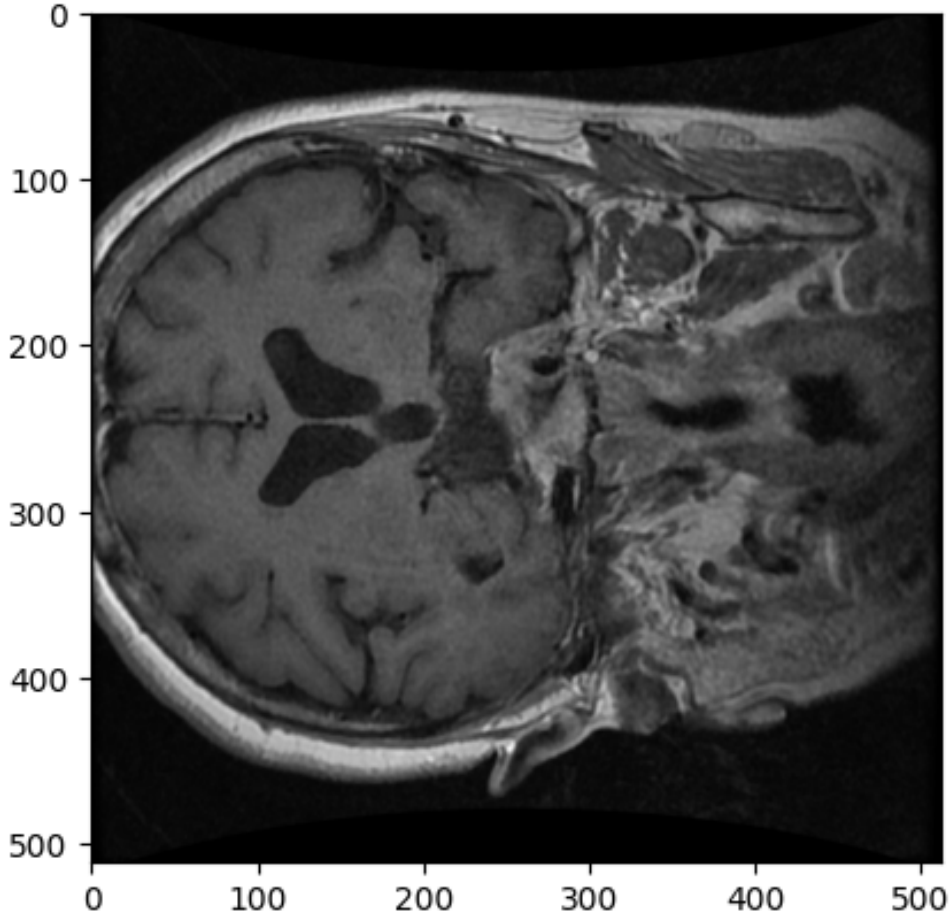
5. M. Buda, A. Saha, and M. A. Mazurowski, "Association of genomic subtypes of lower-grade gliomas with shape features automatically extracted by a deep learning algorithm," *Computers in Biology and Medicine*, 2019.

6. Public architecture and MRI visualization assets used in this report were sourced from: `https://github.com/mateuszbuda/brain-segmentation-pytorch` (assets directory, accessed April 2026).

Appendix A

Extracted Notebook Figure Gallery

All PNG outputs embedded in the notebook were extracted and preserved.



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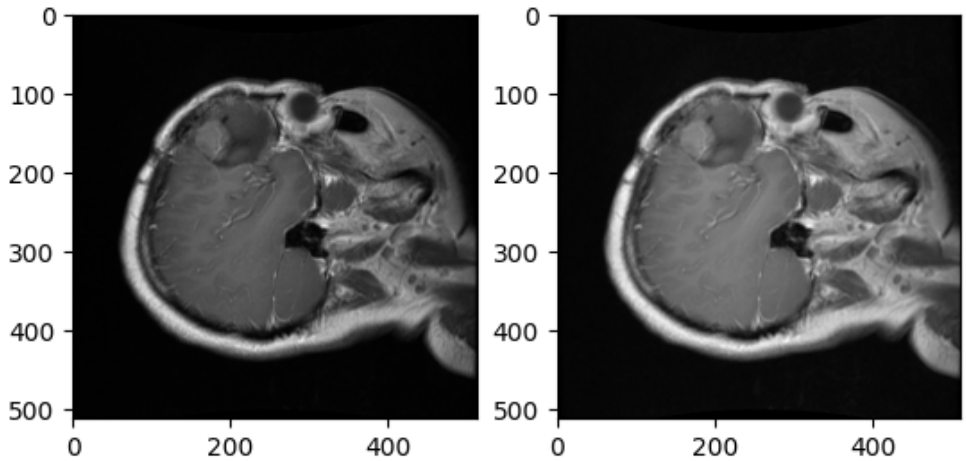

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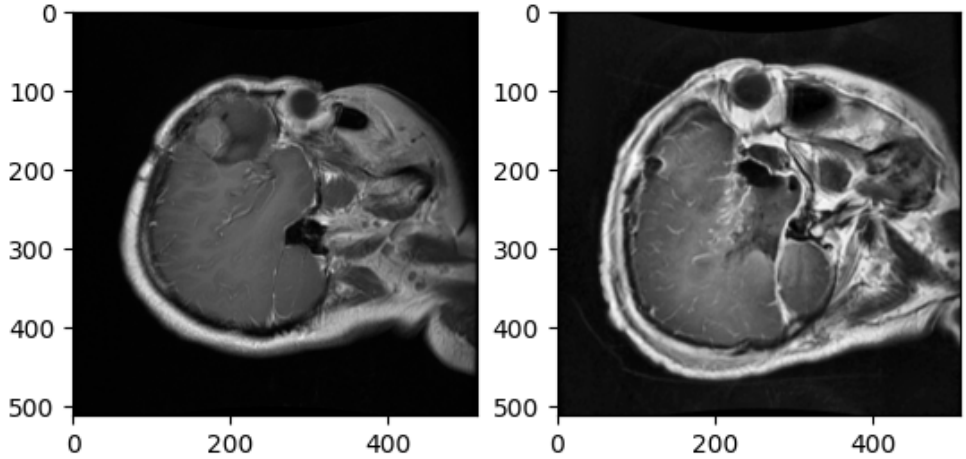

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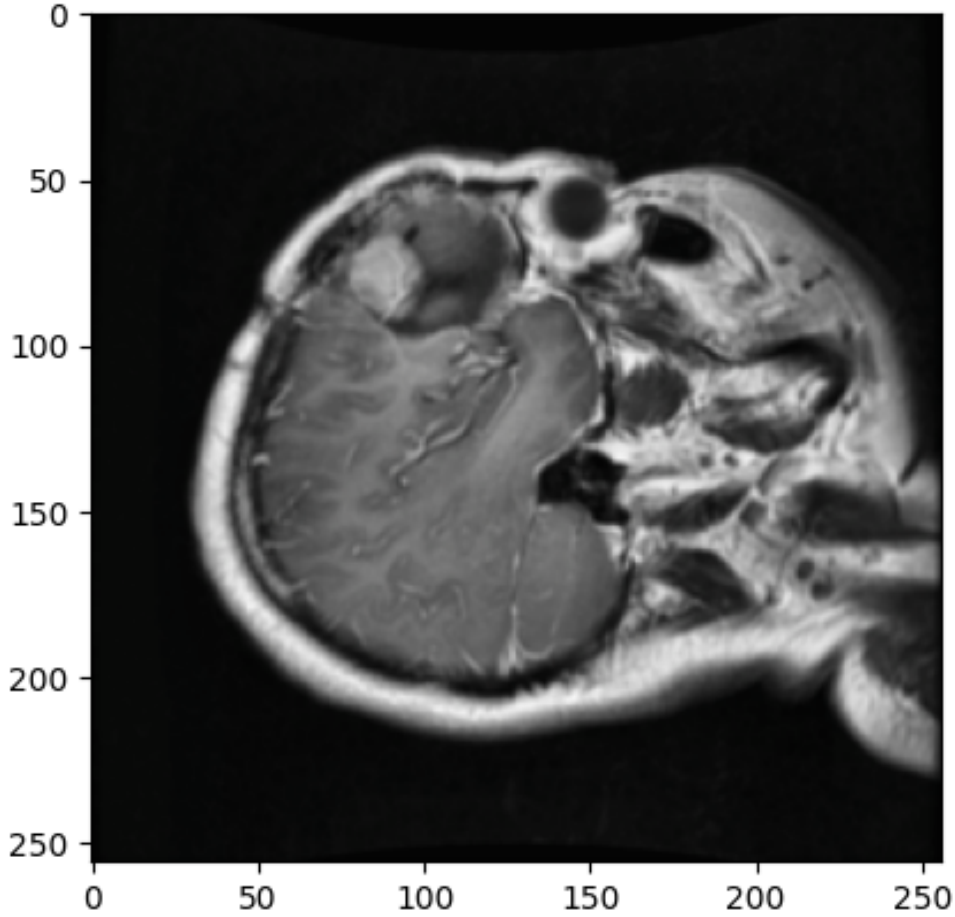
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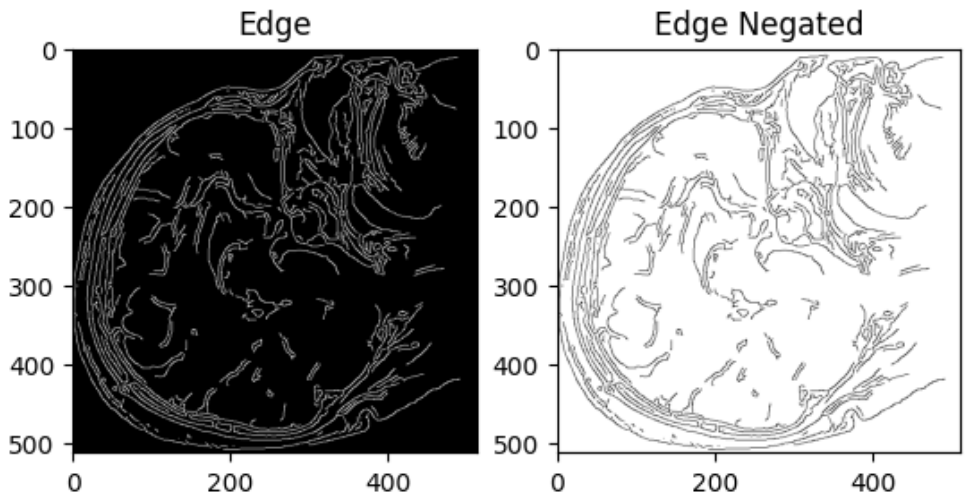
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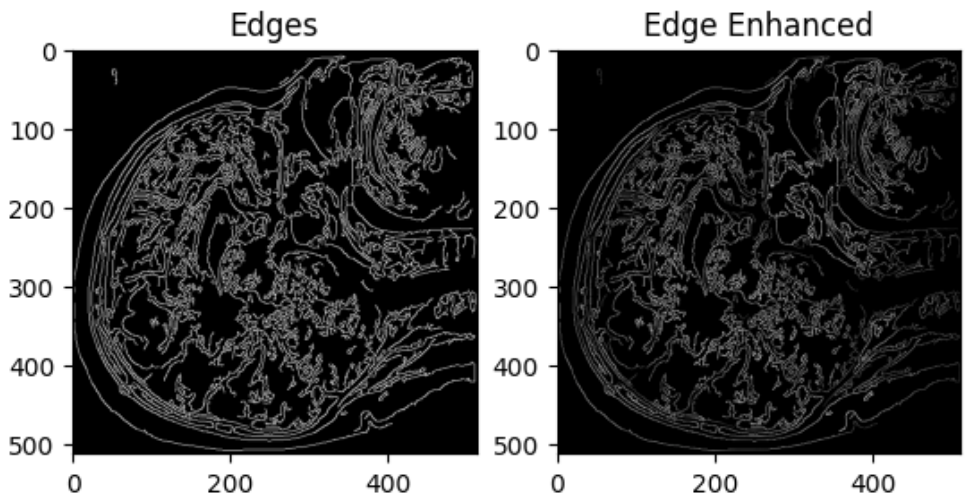
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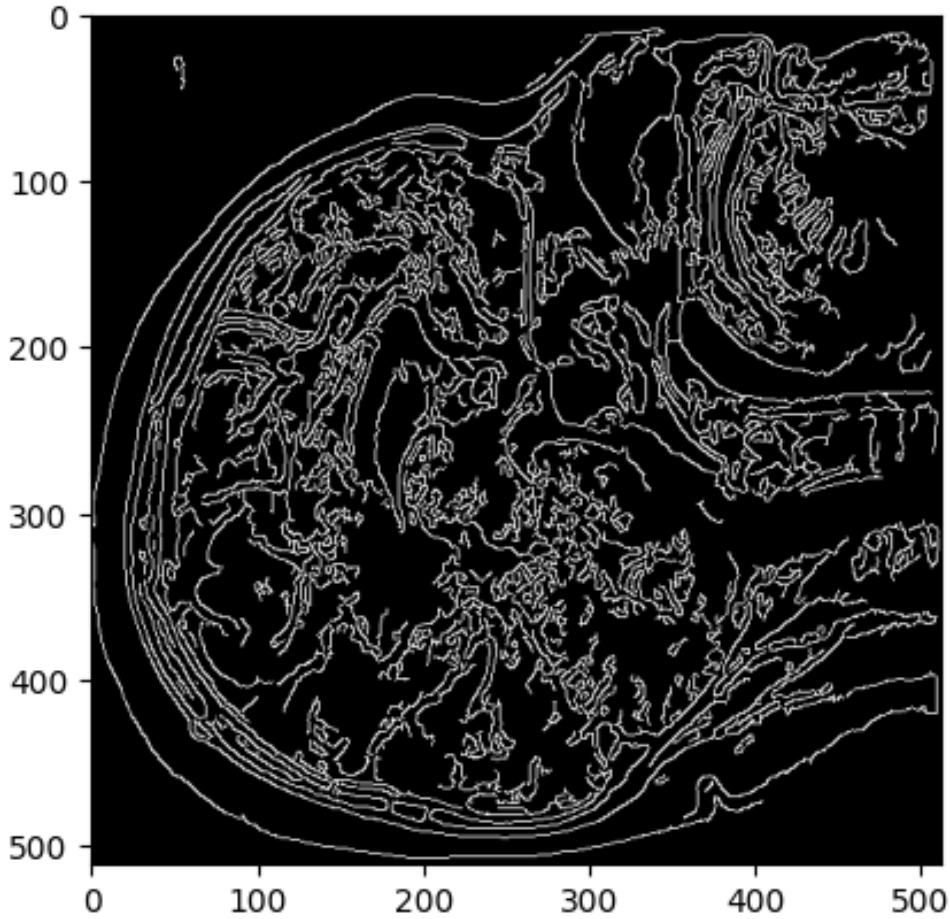
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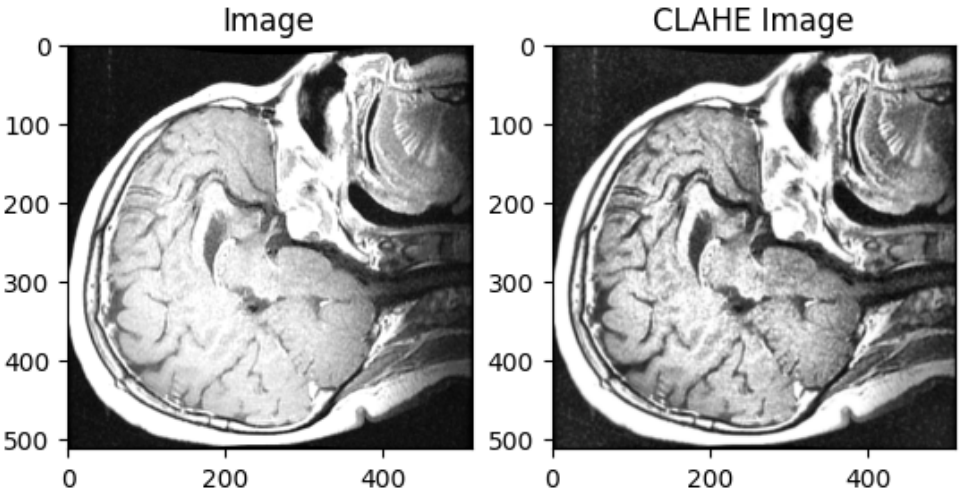
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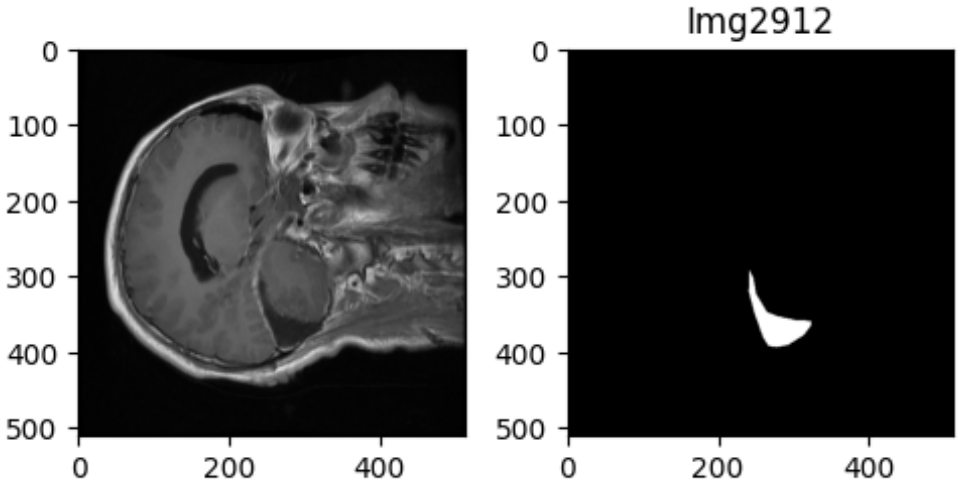
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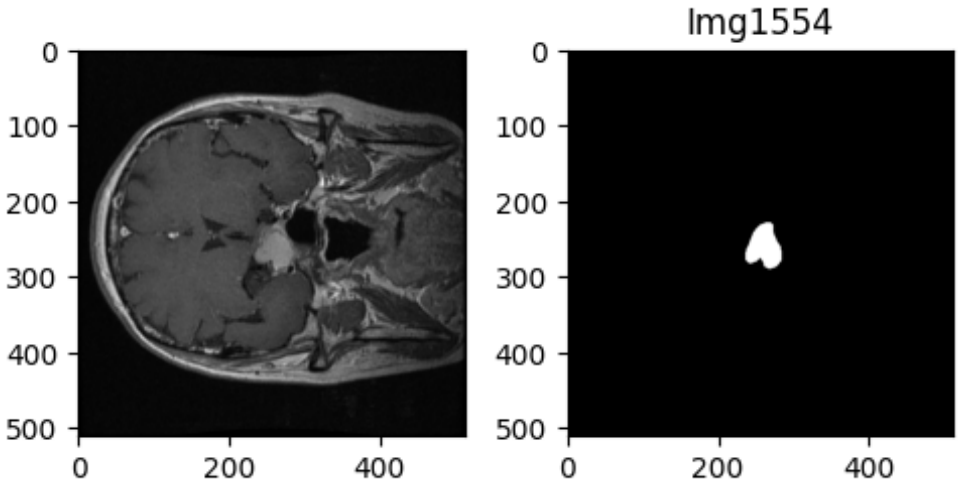
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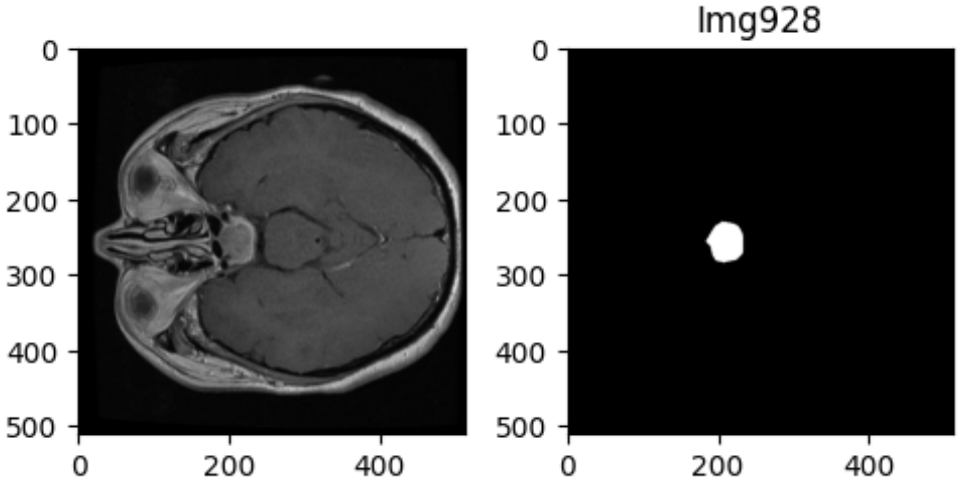
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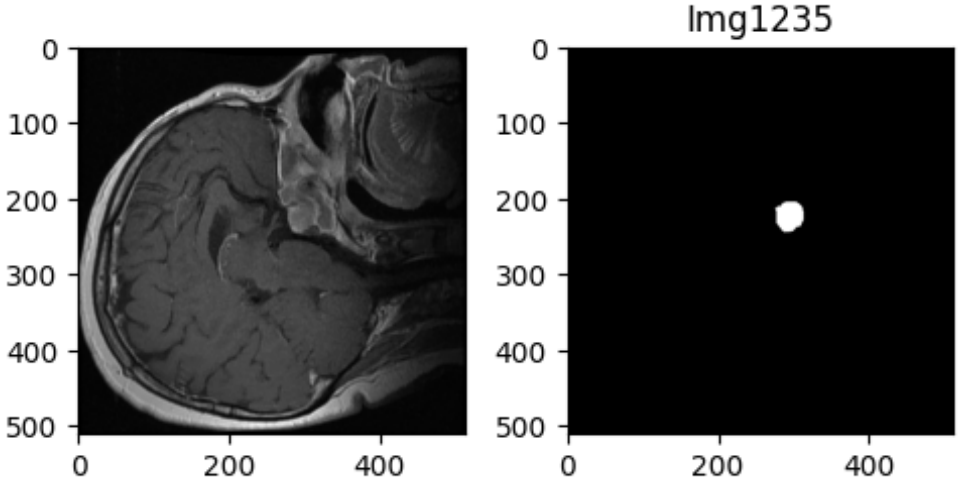
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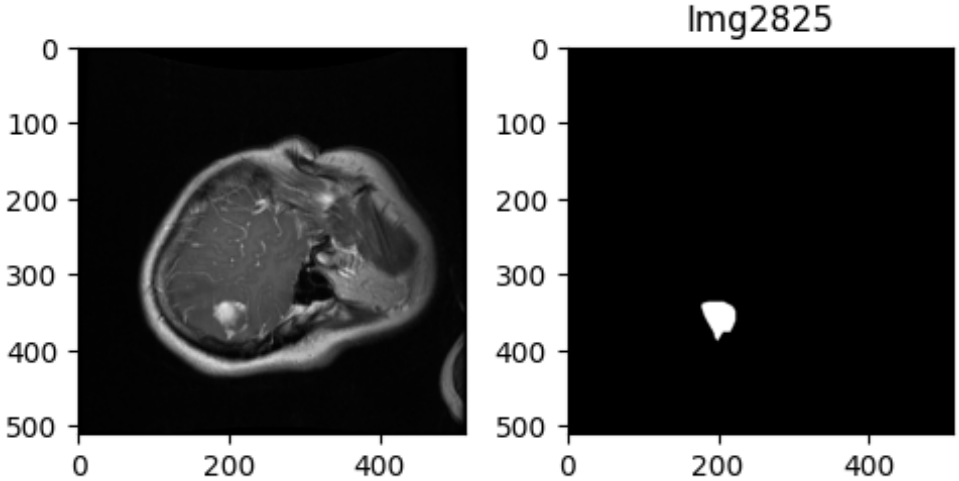
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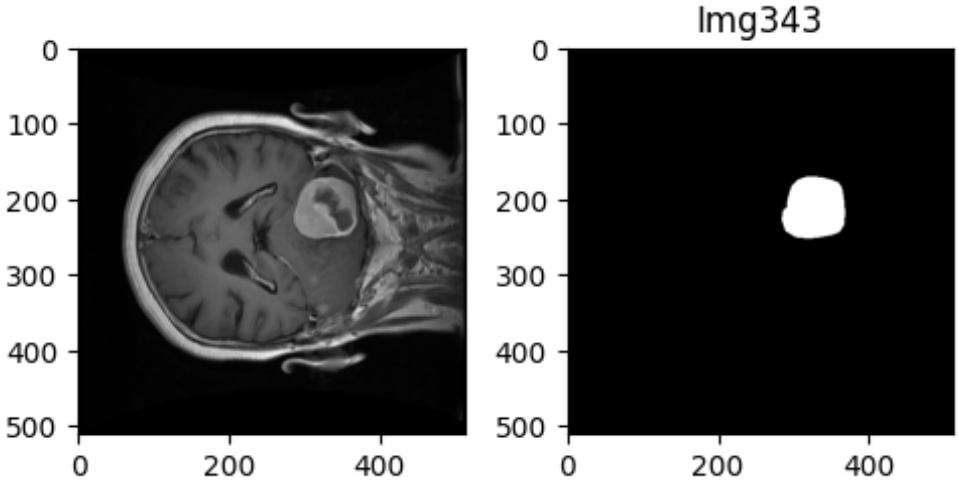
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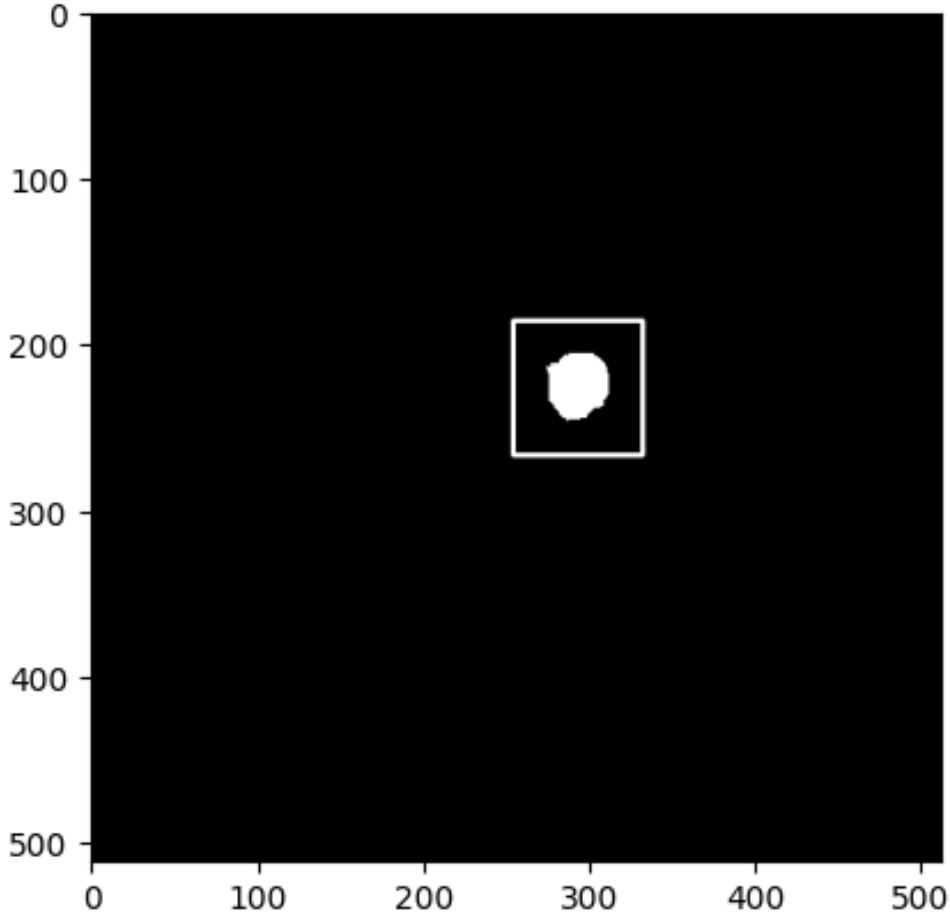
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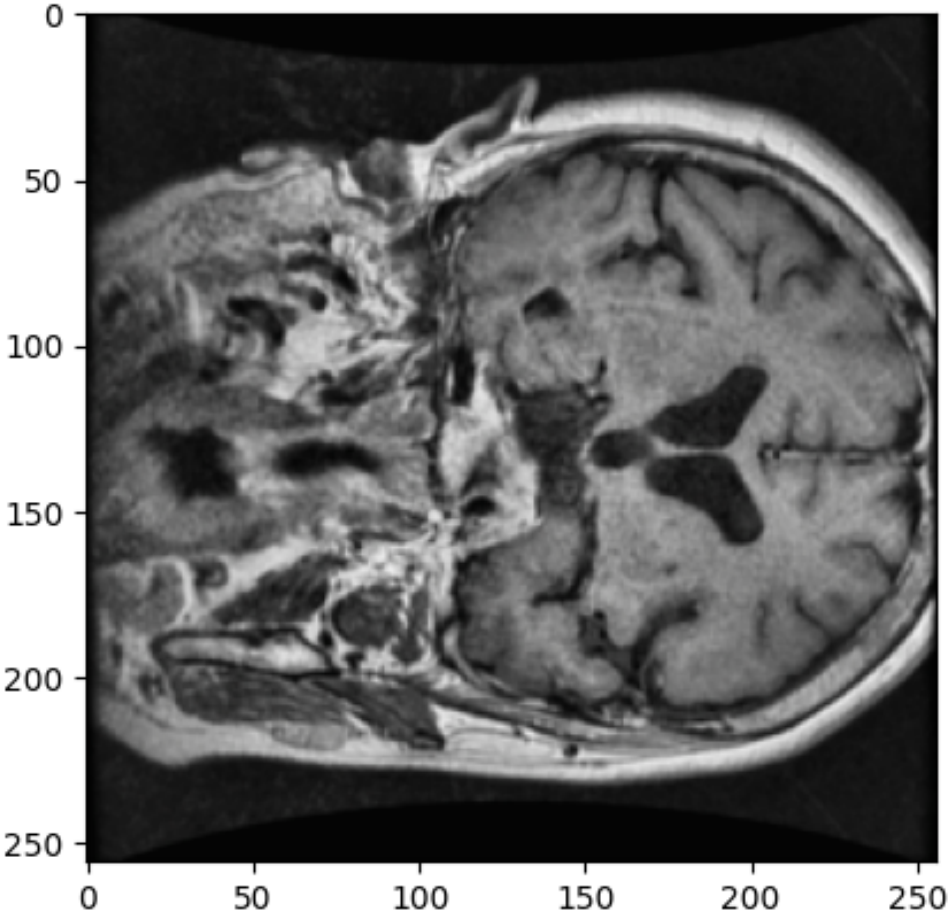
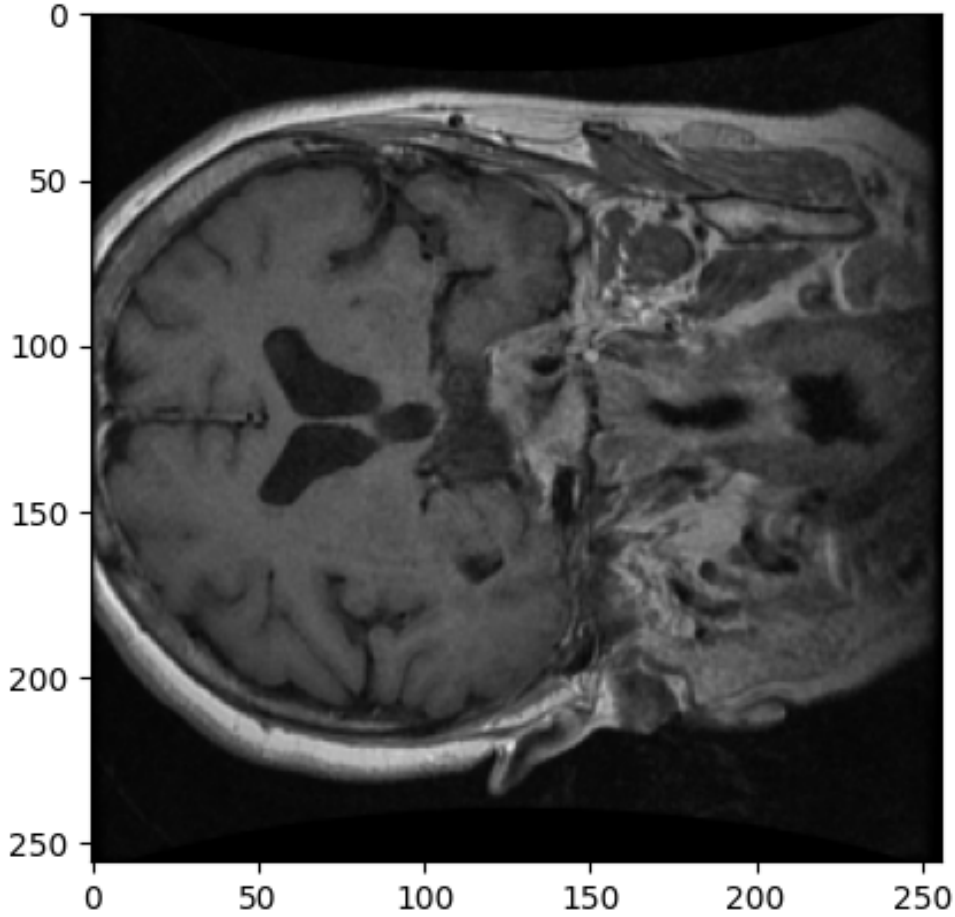


16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100
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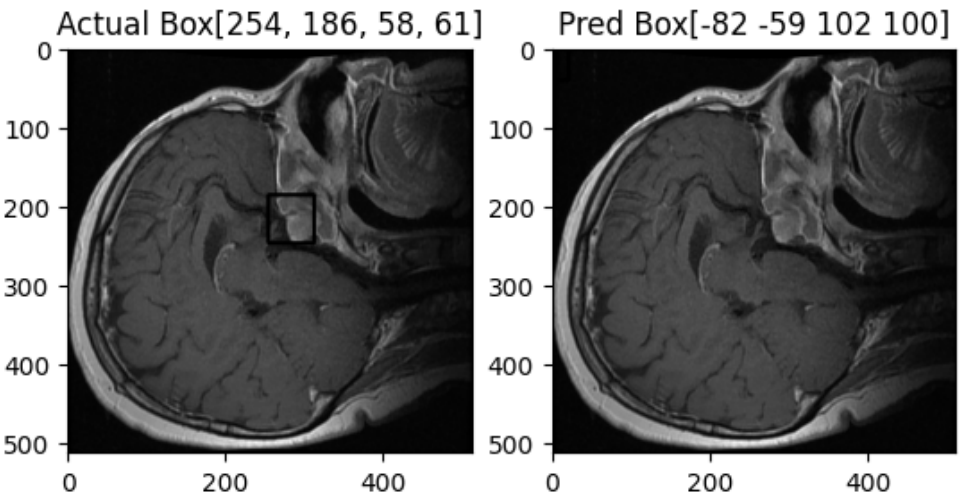
200
250

17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Pred Box[-82 -59 102 100]
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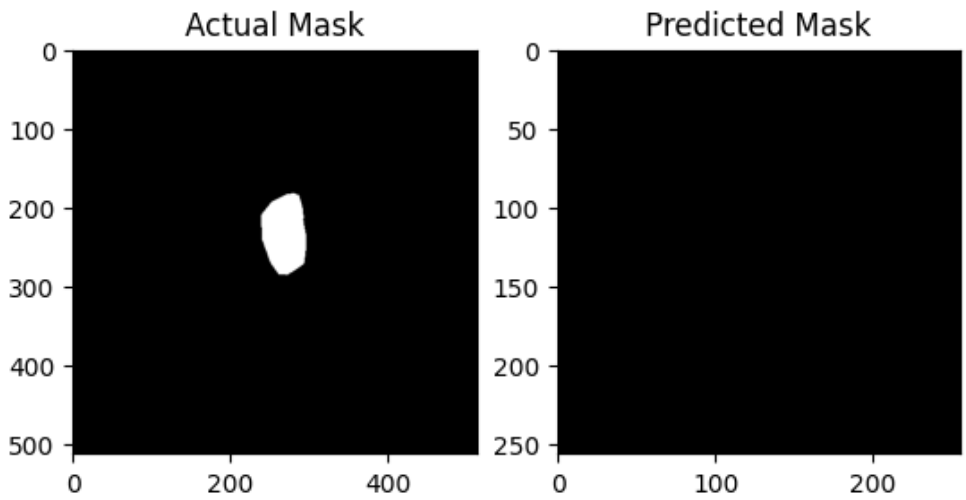
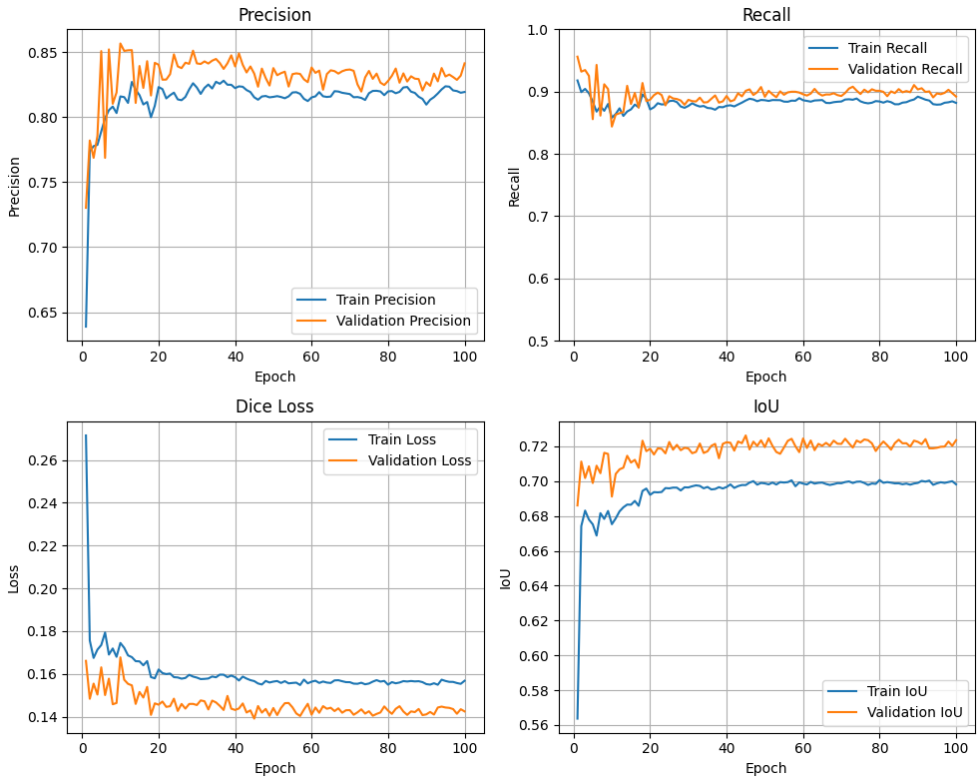
100
200
300
400
500
0
200
400

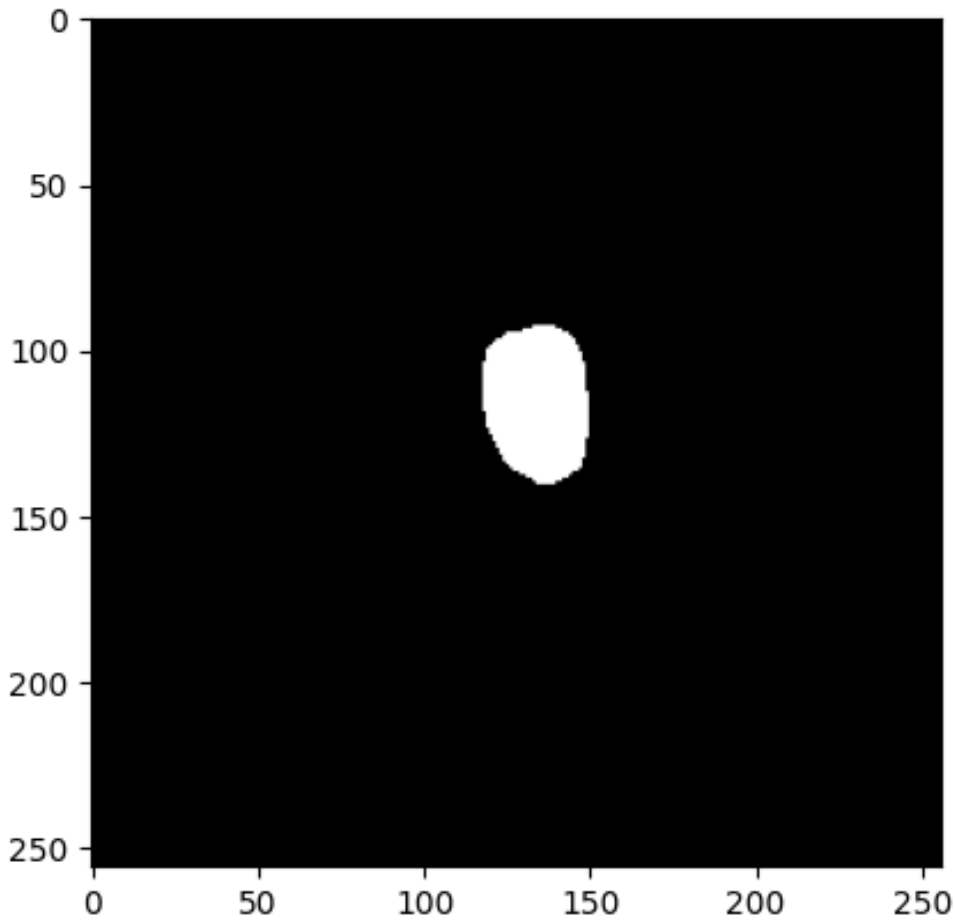
18


Actual Mask
Predicted Mask

19


Precision
Train Precision
Validation Precision
Epoch
Recall
Train Recall
Validation Recall
Epoch
Dice Loss
Train Loss
Validation Loss
Loss
Epoch
IoU
Train IoU
Validation IoU
Epoch

20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150
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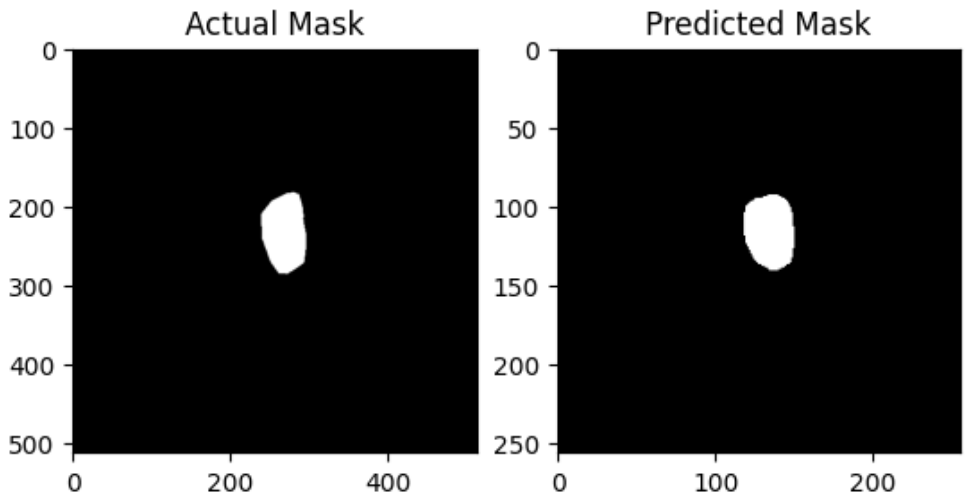
21


Actual Mask
Predicted Mask
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300
400
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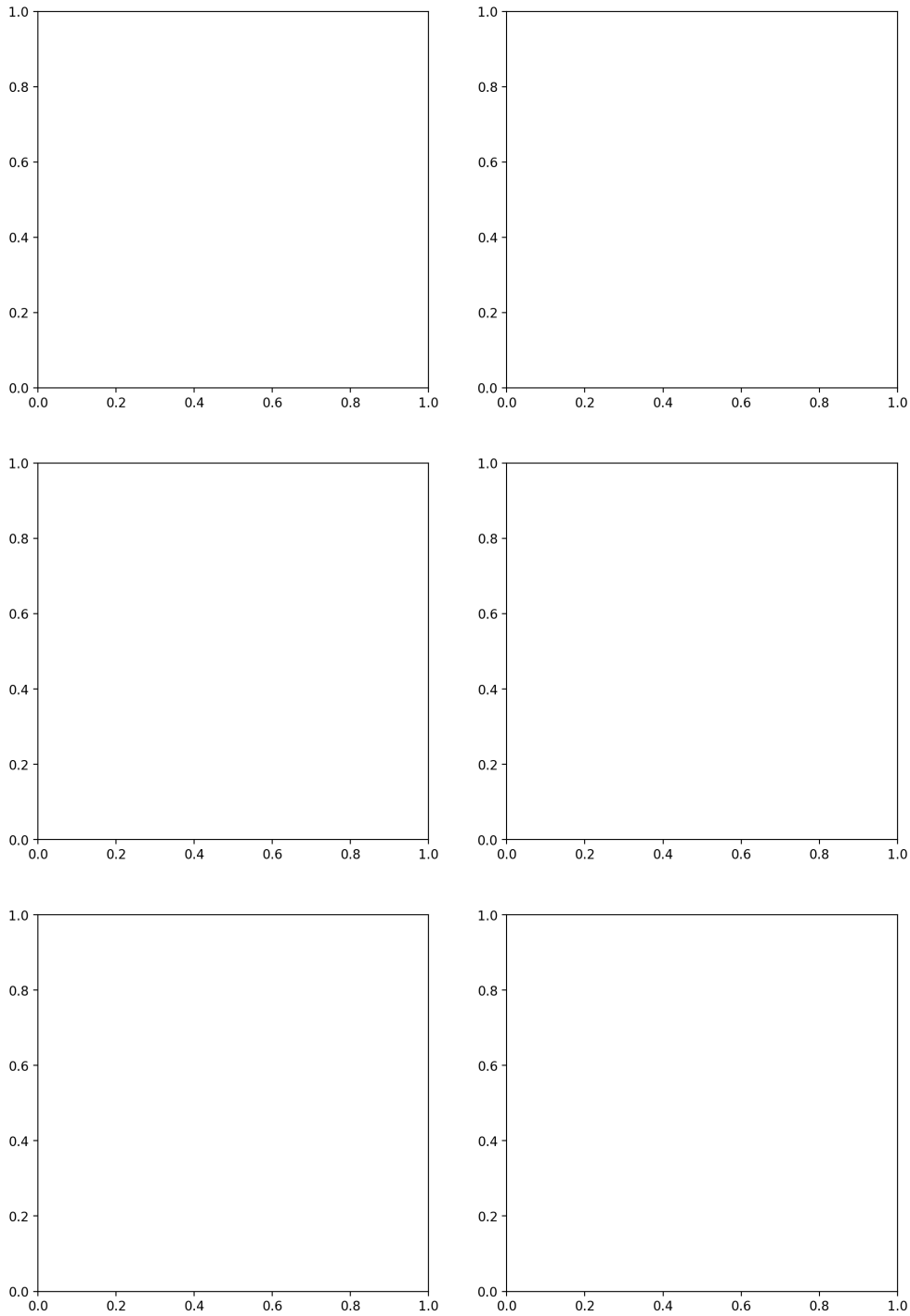
400
0
50
100
150
200
250
0
100
200

22



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